ITAI 3375 — Curating Data for AI Optimization

LAB 09 — Design & Execute an Annotation Pipeline

Module 09: Data Annotation, Labelling & Human-in-the-Loop — Workflow Design, IAA, Active Learning & Programmatic Labelling

SLO Assessed: 2 | Submission Format: This completed document + Label Studio export + notebook

Student Information

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Date Submitted:	07/10/2026	**Section / CRN:**

Lab Overview

This lab maps directly to Module 09 SLO 2:

- SLO 2 — Apply the data curation lifecycle by designing an annotation workflow, measuring interannotator agreement, and applying active learning to improve label efficiency

You will design a complete annotation pipeline end-to-end: define the task and schema, write annotation guidelines, set up Label Studio, annotate 200 items with 3 annotators, compute IAA metrics, implement one active learning round, and produce an annotation quality report.

Part A — Task Definition & Annotation Schema

Objective: Define exactly what is being annotated, why, and how before touching any tools or data.

Section A: Task Definition and Annotation Schema

A1. Choosing a TaskThe chosen task is to classify the sentiment of tweets and short texts into three categories: Positive, Negative, and Neutral (Twitter Sentiment140 or SST-2 style).

Definition of the Task

What is being annotated?

individual brief texts (less than 280 characters, such as tweets or similar communications). Every text has a single sentiment label and is annotated as a single unit.

For what reason is this being annotated?
to produce high-quality labeled data for sentiment analysis models used in opinion analysis, social media monitoring, customer feedback, and brand tracking. This facilitates the lab's requirements for quality reporting, active learning, IAA measurement, and the entire annotation workflow.

In what way is this being annotated?

Type: Multi-class classification using a single label.

Labels: Positive

A tone that is primarily joyful, affirmative, enthusiastic, or hopeful.
Negative: A tone that is primarily critical, depressing, furious, or gloomy.

Neutral: Objective, factual, or devoid of obvious emotional resonance.

Procedure: In accordance with specific standards (to be provided in Part B), three annotators independently label each paragraph depending on overall attitude. Just pay attention to the text that has been presented.

A1. Task Selection

Select ONE annotation task below. All subsequent work applies to this task.

a) **Tweet / short-text sentiment classification — 3 classes: Positive, Negative, Neutral (dataset: Twitter Sentiment140 or SST-2)**
b) News article topic classification — 5 categories: Politics, Technology, Sports, Health, Business (dataset:
AG News or BBC News)
c) Medical symptom NER — entity types: SYMPTOM, BODY_PART, NEGATION (dataset: MedNLI or clinical notes subset)
d) Product review aspect-based sentiment — aspects: Quality, Price, Service, Delivery × Pos/Neg/Neutral (dataset: Amazon Reviews)
e) [ ] My own task (requires instructor approval) — describe task and dataset: ________________

A2. Label Taxonomy

Define every label in your schema. Each label must have a precise, testable definition.

Label / Tag	Precise Definition (testable criterion)	Do NOT label as this if...
Positive	The text expresses clear positive emotions, opinions, or attitudes such as happiness, joy, praise, excitement, approval, optimism, gratitude, or support. The overall tone is favorable.	The majority of the material is impartial, factual, mixed, sardonic, or has any notable negative aspects.
Negative	Anger, grief, frustration, criticism, disappointment, fear, hatred, or pessimism are only a few of the obvious negative feelings, thoughts, or attitudes that are expressed in the text. Overall, the tone is negative.	The text is mostly neutral, factual, mixed, sarcastic, or contains any significant positive elements.
Neutral	The text is factual, objective, informational, or does not express any clear positive or negative sentiment. It may include questions, statements of fact, or balanced/mixed views with no dominant emotion.	he text clearly leans positive or negative in its overall tone.
Mixed	The general tone of the work is ambiguous or contradictory since it contains both strong positive and negative thoughts that are roughly balanced.	The text has a clear dominant positive or negative tone.

A3. Annotation Guidelines Document

Write your complete annotation guideline in the space below. Minimum required sections: (1) Task description, (2) Label definitions, (3) Decision tree or flowchart, (4) Three positive examples per label, (5) Three negative examples per label, (6) Escalation protocol for ambiguous cases.

Label Definitions

- **Positive**: The text expresses clear positive emotions or opinions example: happiness, praise, excitement, approval, optimism, gratitude etc.
- **Negative**: The text expresses clear negative emotions or opinions example: anger, sadness, criticism, disappointment, fear, hate, pessimism, etc.
- **Neutral**: The text is factual, objective, or does not show a clear positive or negative sentiment.

Task Description

Annotators will label short texts (tweets or similar messages) with one of three sentiment classes: Positive, Negative, or Neutral. The goal is to determine the overall dominant sentiment of the text. Focus only on the provided text. Ignore usernames, timestamps, images, or external context.

Decision Tree / Key Decision Points

Describe in plain language the decision sequence annotators should follow. Attach a drawn flowchart if possible.

Read the entire text carefully.

Ask: Does the text have a **clear positive tone** overall? → **Positive**

Else: Does the text have a **clear negative tone** overall? → **Negative**

Else (no clear dominant emotion, factual, or balanced) → **Neutral**

Just got my dream job! Feeling so grateful and excited for this new chapter

This new phone is absolutely amazing! Best purchase ever.

What a beautiful day! Love spending time with family

Negative Examples:
Worst customer service ever. So disappointed and frustrated. This movie was terrible. Waste of time and money.I hate how expensive everything is getting. This is ridiculous.
Examples that are neutral:

The meeting is set for tomorrow at 3 PM.
Today, Apple published a new software update.
Rain is expected this weekend, according to the weather prediction.

5. Each label has three negative examples (counter-examples).

Negative: The service was slow, but the food was passable. Negative or Neutral
Neutral: This is the best day ever! Not Negative: I didn't enjoy it (too unclear without stronger tone)
→ Consider Neutral Positive

Positive Examples (3 per label minimum)

#	Text / Item	Correct Label	Why this label
1	I recently landed the job of my dreams! I am so appreciative and eager for this new phase Positive conveys joy, appreciation, and enthusiasm	Positive	Express Hapiness, gratitude and excitement
2	2. This new phone is fantastic! The best thing ever bought.Positive Strong commendation and endorsement	Positive	Strong Praise
3	What a lovely day! I enjoy spending time with my family. Positive, joyful, and loving	Positive	Love and clear joy
4	4 Wow! All of my expectations were surpassed by this eatery. Strongly endorsed and highly recommended!	Positive	Recommendation and Strong approval
5	5 The quality of this film is astounding. I adored it!Strong good feelings and enthusiasm	Positive	Strong positive emotion
6	I really appreciate your assistance! You're the greatest! Positive appreciation and commendation	Positive	Gratitude and praise
7	7 I've finally attained my fitness objective. Feeling unstoppable!	Positive	Achievement and positive self feeling

	Positive self-perception and accomplishment		
8	8 I have never received a better gift than this one. Very considerate!	Positive	Joy and appreciation
9	Incredible performance tonight! So proud of the team!	Positive	Pride and excitement

Negative Examples / Common Mistakes (3 per label minimum)

#	Text / Item	Mistake: Labelled As	Correct Label + Reason
1	I've never purchased a worse product than this one.	Positive	Negative: Disappointment and harsh criticism
2	I hate waiting in long lines at the airport	Positive	Negative – Expresses clear frustration
3	The customer assistance was incredibly unhelpful and delayed.Neutral Negative:	Neutral	A specific grievance over the service Strong Negative Emotion
4	My phone keeps crashing. Terrible experience. Neutral	Neutral	
5	Not sure if I like the new update	Positive	Neutral, Mixied and unclear
6	Everything went according to plan during the meeting.	Negative	Neutral – Purely factual statement
7	This movie was okay but the ending was bad.		Negative

8	This necklace cute but the price is too high	Neutral	Conflicting, Negative and Mixed
9			

Part B — Label Studio Setup & Annotation

Objective: Configure Label Studio, annotate 200 items independently with 3 annotators, and export the results.

B1. Label Studio Configuration

Deployment method	☐ **Docker** local ☐ Label Studio Cloud ☐ HuggingFace Spaces ☐ Other
Project name	Tweet Sentiment Classification 3 Class
Label config type	☐ **Text Classification** ☐ NER ☐ Image ☐ Custom XML
Paste first 3 lines of label config XML	<View> <Text name="text" value="$text"/> <Choices name="sentiment" toName="text" choice="single" showInline="true"> <Choice value="Positive"/> <Choice value="Negative"/> <Choice value="Neutral"/> </Choices> </View>
Gold set size	Min 30 items — how many did you create?
Gold set source	How did you determine the correct labels for gold items?

	I carefully chose and identified the gold set pieces. I selected illustrative instances that encompassed a variety of unambiguous and unclear situations. The right labels were identified by: thorough manual assessment based on the official annotation rules. cross-referencing with labeled datasets that are already available like Sentiment140. High trust in the ground truth labels is ensured by consensus from several evaluations. Later on in the process, annotator accuracy and agreement will be assessed using these gold pieces.

B2. Annotator Information

Annotator	Background / Domain Knowledge	Training Completed?
A1	Computer Science student with experience in NLP and social media analysi	Yes
A3	Majoring in linguistics and having experience with sentiment and text annotation jobs	Yes
A3	Data Science student, some prior experience with labeling tasks	Yes

B3. Calibration Results (Gold Set)

Each annotator must achieve ≥ 80% accuracy on the 30 gold items before labelling the full dataset.

Annotator	Gold Items Correct	Gold Accuracy %	Pass (≥80%)?
A1	46/50	92%	Yes
A2	44/50	88%	Yes
A3	43/50	86%	Yes

If any annotator failed calibration, what retraining steps were taken before proceeding?

What retraining procedures were followed before moving forward if any annotator failed calibration?

With accuracies of 92%, 88%, and 86% on the 50 gold objects, respectively, all three annotators A1, A2, and A3 passed the calibration stage on their first try. There was no need for retraining.

But just to be safe, all annotators:
Together, I went over the entire annotation guidelines document. I
talked about ambiguous examples and edge circumstances (such as sarcasm and mixed sentiment).
Before beginning the primary 200-item annotation, I finished a brief practice round of ten more things with feedback.
Before moving on to the complete dataset annotation, this guaranteed great consistency and alignment.

B4. Annotation Export

Total items annotated	Must be 200 (+ 50 gold set) = 250 total annotaded items
Export format	☐ JSON ☐ **CSV** ☐ CONLL (NER tasks) ☐ Other
Annotator ID field name	Column that identifies which annotator labelled each item annotator_id
Label distribution	Label 1: P 82 | Label 2: N 68 | Label 3: N 50...

Part C — Inter-Annotator Agreement (IAA)

Objective: Compute Cohen's κ for each annotator pair and Fleiss' κ across all three. Identify the label category with the lowest agreement.

C1. Pairwise Cohen's κ

Annotator Pair	Observed Agreement P_o	Cohen's κ	Interpretation

A1 vs. A2	0.87	0.80	Strong agreement
A1 vs. A3	O.84	0.76	STRONG AGREEMENT
A2 vs. A3	0.82	0.73	Subtantial agreement

Fleiss' κ (all 3 annotators): κ_F = ____all three annotators___________ Interpretation: __0.76_____________

C2. Per-Label Agreement Breakdown

Which label has the lowest pairwise agreement? Compute the percentage of times each pair agreed on each label.

Label	A1–A2 Agreement %	A1–A3 Agreement %	A2–A3 Agreement %
Positive	91%	88%	86%
Negative	85	82	80
Neutral	78%	74%	72%

Lowest-agreement label: ___________Neutral____ Mean agreement on this label: _75%__%

Why does this label have lower agreement? What ambiguity in the guidelines is causing it?

Due to its subjectivity, the Neutral classification has the lowest level of agreement. Ambiguities occur when:

Subtle irony
The single tweet contained both good and negative feelings.
Brief facts that, depending on the annotator's viewpoint, could be perceived as somewhat good or negative
More examples of borderline neutral cases as opposed to marginally positive or negative cases would enhance the rules.

C3. Most Common Disagreement Pattern

List the top 3 annotator disagreements by type (e.g., 'A1 labels X as POS, A2 labels same as NEU'). For each, propose a specific guideline update that would resolve it.

#	Disagreement Pattern	Frequency	Proposed Guideline Update
1	A1 classifies tweets as favorable, whereas A2 and A3 classify them as neutral (mildly positive).	12	Add clearer rule: If the positive language is weak or qualified (e.g. “pretty good”, “not bad”), default to Neutral unless strong positive words/emojis are present.
2	A2 labels as Neutral, A1/A3 label as Negative (subtle complaints or sarcasm)	9	Include particular instances of subtle grievances and sarcasm. Include rule: Even if the phrase form is courteous, look for derogatory words.
3	Mixed sentiment tweets (both Pos & Neg in one text)	7	Make a new section on mixed emotions guidelines. Unless there is a clear dominance of one sentiment, it is advised to identify as neutral.

Part D — Active Learning Round

Objective: Train a baseline classifier, apply entropy sampling to select the 20 most uncertain items, label them, retrain, and measure accuracy improvement.

D1. Baseline Classifier

Model architecture	e.g., **Logistic Regression on TF-IDF,** BERT-tiny, distilbert-base-uncased
Training set size	80% of 200 items = ___160__ items
Held-out test set size	20% of 200 items = __40___ items
Features / tokenisation	e.g., **scikit-learn TfidfVectorizer**, HuggingFace AutoTokenizer with English stop word removed, ngram_range-1,2
Baseline accuracy	**On held-out** 76.6% 20% — must be > random chance
Macro-averaged F1	Per-class and overall 0.74

D2. Entropy Sampling

Apply entropy sampling to the 20% held-out set (or a new unlabelled pool). Compute prediction entropy for each item and select the 20 with the highest entropy.

Entropy formula used: $H(x) = -\Sigma P(y□|x) \cdot \log_2 P(y□|x)$

Entropy computed using	e.g.**, model.predict_proba()** from scikit-learn, (Logic Regression)

Pool of items sampled from	How many items was entropy computed for? For the purpose of active learning, the 40-item held-out test set is unlabeled.
Highest entropy item	Show the item text and its predicted probability distribution Text: “This is interesting, but I’m not sure if it’s good or bad tbh.” Predicted probabilities: Positive: 0.38, Negative: 0.35, Neutral: 0.27 Entropy: 1.58 (very high uncertainty)
Mean entropy in top-20 selected	vs. mean entropy of full pool 1.42 mean entropy of full pool: 0.89
Do top-20 cluster around a particular label?	e.g., mostly boundary between POS and NEU? Indeed, most high-entropy objects fall between the Positive and Neutral boundaries, with a small number falling between the Negative and Neutral boundaries. There were very few highly negative high-entropy things.

Paste entropy distribution summary (top-5 most uncertain items with their P(y|x) distributions):

```
Text:This is intriguing, but to be honest, I'm not sure if it's good or bad.
Entropy: 1.58  Prob: Positive 0.38 Negative 0.35  Neutral 0.27
Text: "It's a little slow now, but I kind of liked the new update.
Entropy: 1.55  Prob: Positive 0.42  Negative 0.33  Neutral 0.25
Text: I suppose the service was alright.
Entropy: 1.52  Prob: Positive 0.45 Negative 0.28 Neutral 0.27
Text: It wasn't the greatest experience, but it wasn't the worst either.
Entropy: 1.51 Prob: Positive 0.36 Negative 0.39 Neutral 0.25
Text: Well, it wasn't too bad.
Entropy: 1.49 Prob:
Positive 0.31
Negative 0.30
Neutral 0.39
```

D3. Label the 20 Uncertain Items

Who labelled the 20 items?	☐ **Single expert annotator** ☐ Majority vote of all 3 annotators ☐ **Single annotator + gold check**
IAA on these 20 items	Cohen's κ if multiple annotators labelled — expected to be lower than full dataset Not applicable Single Expert Label
Label distribution of 20 new items	Label 1: p: 7 | Label 2: N :6| Label 3: N :7...
Were any items found to be mislabelled in the original set?	☐ **Yes** —3 how many? ☐ No

D4. Retrained Model Comparison

Metric	Baseline (160 items)	After AL Round (180 items)
Overall Accuracy	76.5%	81%
Macro-averaged F1	.74	.79
F1 — Label 1 (Positive it)	.78	.83
F1 — Label 2 (name it) Negative	.72	.76
F1 — Label 3 (name it) Neutral	.68	.74

Did active learning improve performance? For which labels did it help most? Why?

Indeed, performance was enhanced by active learning.
Macro F1 increased by +0.05 and overall accuracy by +4.5%.

Neutral benefited the most from it (F1 increased from 0.68 to 0.74). This makes sense given the majority of the uncertain (high-entropy) elements involved Neutral border instances. The model learned improved decision boundaries for ambiguous scenarios by incorporating these challenging examples.

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Part E — Annotation Quality Report

Objective: Produce a professional annotation quality report summarising all findings, guideline updates, and recommendations.

E1. Executive Summary (1 paragraph)

State: the task annotated, IAA result and what it means, the most problematic label category, active learning improvement, and the single most impactful guideline change needed.

E2. Guideline Update Specification

Specify 3 concrete updates to your annotation guidelines based on the disagreements you observed. Each update must reference a specific confusion pattern.

#	Confusion Pattern Addressed	Specific Guideline Text to Add / Change
1	Mildly positive language labeled as Neutral or Positive	Add the following rule: "Label as Neutral if the text contains weak positive phrases (e.g., "pretty nice," "okay," and "not awful") without strong positive words or emojis."
2	Subtle complaints / sarcasm confused with Neutra	"Sarcasm and subtle complaints should be labeled Negative when negative words or tone are present, even if the structure is polite," should be included.
3		
#	Confusion Pattern Addressed	Specific Guideline Text to Add / Change
	Mixed sentiment tweets both Positive & Neggative in one text	"Label texts with obviously conflicting sentiments as neutral unless one sentiment clearly dominates the entire message."

E3. If You Were to Annotate 100K Items — What Would Change?

Reflect on scaling: which parts of your pipeline would break or become infeasible at 100K items? What tools, infrastructure, or quality-control approaches would you add?

With just three annotators, some aspects of the current pipeline would become impractical above 100K items. Annotating by hand would be too time-consuming and costly.

Important adjustments I would make:
From the beginning, switch to a hybrid human-in-the-loop system that combines semi-supervised and active learning methods.
Only uncertain or high-entropy items should be reviewed by humans when using pre-labeling with a robust model (such as refined BERT).
Increase the number of gold questions, identify low-agreement items automatically, and create annotator performance dashboards to implement quality control at scale.
Improve the infrastructure: For distributed annotation with appropriate work assignment and payment monitoring, use Label Studio or Scale AI/Surge AI.
Incorporate bias monitoring, low-effort annotations, and automated spam checks.

E4. Ethical Reflection

If this annotation task involved sensitive content (political opinion, emotional distress, medical information, NSFW material), what additional ethical safeguards would you put in place for the labelling workforce?

I would put in place the following safeguards for the labeling workers if this activity entailed sensitive content (such as political beliefs, emotional anguish, hate speech, or medical information):

Provide services for mental health and permit annotators to exclude upsetting content without facing consequences.
Use material warnings and trauma-informed training prior to assignments.
Assure equitable pay and sensible workload restrictions for each shift.
Preserve annotators' privacy and safeguard their information.
Keep an eye out for bias in labeling choices and burnout.
Establish explicit procedures for reporting dangerous content and open lines of communication regarding the data's intended purpose.

Grading Rubric

Component	Criteria for Full Credit	Points
Part A — Task & Guidelines	Label taxonomy is complete and testable; guideline includes decision tree and ≥ 3 positive + negative examples per label; escalation protocol defined	20
Part B — Label Studio & Annotation	Calibration documented with gold accuracy per annotator; all 3 annotators passed threshold; 200-item export confirmed with label distribution	15
Part C — IAA Computation	Pairwise Cohen's κ computed for all 3 pairs; Fleiss' κ computed; per-label breakdown complete; top-3 disagreement patterns identified with proposed fixes	25
Part D — Active Learning	Baseline accuracy recorded; entropy sampling implemented and top-20 selected; 20 items labelled; retrained model comparison table complete with analysis	25
Part E — Quality Report	Executive summary is clear and specific; 3 concrete guideline updates reference actual disagreement patterns; scaling and ethical reflections are substantive	15
Total		**100**

Instructor feedback: